

RIPHAH INTERNATIONAL **UNIVERSITY, ISLAMABAD**



Bachelors of Computer Science – BS-CS6

Course: Advanced Computer Programming

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Lost & Found Web Application (MERN Stack)

1. Project Overview

The **Lost & Found** web application is a full-stack MERN-based platform that allows university students and faculty to report lost items and view found items. This system bridges the communication gap between item reporters and finders. It incorporates **secure user authentication, item claim workflows, and admin-side verification capabilities.**

2. Objectives

- Allow users to report and search for lost or found items.
- Store user reports securely using JWT-based authentication.
- Protect sensitive APIs using Express middleware.
- Provide admins with structured, verifiable submissions for faster recovery.
- Ensure usability through a responsive UI (React with Vite).

3. Main Features & Functionalities

3.1 Home Page

- Hero section with CTA buttons for "Report Lost" and "Found Items".
- Sections describing how the platform works.

3.2 Found Items

- Grid of found items with images, titles, and claim buttons.
- Clicking "Claim" opens a form to submit claim with reasoning.

3.3 Report Lost Item

- Form to report lost items: includes fields like name, SAP ID, date, description, and category.
- Integrated with MongoDB and Axios for data persistence.

3.4 Claim Item Form

- Form asks for name, SAP ID, department, reason for claim, etc.
- Data sent to the backend for admin review.

3.5 Authentication Screens

- Login and Signup screens implemented.
- JWT token stored and verified for protected routes.

4. Libraries & Technologies Used

4.1 Frontend

Library/Tech	Purpose
React + Vite	Fast modern frontend framework
Axios	API integration with backend
React Route DOM	SPA navigation and routing
Framer Motion	Animations (hero images, buttons, etc.)
React Icons	UI icons for features

4.2 Backend / Database

Library/Tech	Purpose
Node.js + Express	Backend server and routing logic
MongoDB + Mongoose	Data storage and schema validation
bcryptjs	Password hashing
jsonwebtoken (JWT)	Authentication tokens
dotenv	Environment configuration

5. Website Architecture

- **Frontend** uses component-based structure (pages/, components/).
- **Backend** is modularized (routes/, models/, middleware/, controllers/).
- JWT-based authentication secures routes.
- Claims and lost item data stored in MongoDB via Mongoose models.
- All API requests routed through axios with error handling.

6. User Experience Enhancements

- Responsive design using Flexbox and Grid.
- Focused forms with validation and input labels.
- Clean card-based UI to make items easily scannable.
- Smooth page transitions using animations.
- Auto-centered modals and form feedback alerts.

7. Future Enhancements

- Admin dashboard to manage and approve claims.
- Email notifications on matched lost items.
- Advanced search and category-based filtering.
- Pagination for long item lists.
- Deployment to render.com or Vercel.

8. Conclusion

This project successfully implements a real-world problem using modern web technologies. It applies the MERN stack with JWT authentication, protected APIs, dynamic UI, and database integration. The codebase is scalable and ready for deployment with role-based enhancements and user-specific dashboards.